

transactions being appended to Bitcoin's blockchain. As a reward, that miner gets paid in a coinbase transaction, which is the first transaction in the block. Currently, that transaction delivers 12.5 bitcoin to the lucky miner.

The computers involved in Bitcoin's mining process take four pieces of data: a hash of the transactions for that block, the hash (identifier) of the previous block,³ the time, and a random number called the *nonce*. Different computers on the network take these four variables and increment the nonce, perhaps starting with a nonce equal to 0, then going to 1, then to 2, hoping that by changing this one variable the hash output will meet the necessary requirement of the number of starting zeros. The more nonces the miner can test, the more chances the miner will find a golden hash that meets the requirement. The rate at which new nonces can be tested is called the *hash rate*; it is the number of times per second a computer can run these four variables through a hash function and derive a new hash.

Anyone with a computer can connect to Bitcoin's network, download past blocks, keep track of new transactions, and crunch the necessary data in pursuit of the golden hash. Such open architecture is one of Bitcoin's strongest points. While that might sound like an easy way to earn bitcoin, it is now incredibly difficult. Since the launch of Bitcoin, not only have the number of computers mining it increased, but the types of computers used have evolved significantly.

Initially, computers on the network crunched through hashes using their central processing unit (CPU), which is the